

Air Quality, Pollution & Climate Analytics Dashboard

Interactive Exploratory Data Analysis using Python & Streamlit

Prepared by:

Nasim Anwar
(12400578)

Training Organization: O7 Services, Jalandhar

GitHub Repository: https://github.com/Nasim-007-git/AQI_ANALYSIS_O7

1. Introduction

Air Quality, Pollution & Climate Analytics is an interactive data analytics dashboard developed using Python and Streamlit to analyze historical air quality, pollution, and climate data. The dashboard enables users to explore environmental trends through interactive visualizations and statistical analysis. It provides insights into Air Quality Index (AQI), pollutant concentrations, weather conditions, and climate variables across different cities.

The project combines Exploratory Data Analysis (EDA), interactive dashboards, and CPCB-based AQI calculations into a single platform. Users can analyze pollution trends, compare multiple cities, identify pollutant distributions, and calculate AQI using the Central Pollution Control Board (CPCB) standards. The application provides an easy-to-use interface for researchers, students, and environmental analysts to make informed decisions based on environmental data.

2. Problem Statement

Air pollution has become one of the most serious environmental challenges affecting public health and climate. Although large volumes of air quality data are available, extracting meaningful insights using spreadsheets or static reports is difficult and time-consuming.

Environmental researchers and decision-makers require an interactive system that can:

- Analyze AQI trends over time.
- Compare pollution levels across different cities.
- Study pollutant distributions.
- Visualize environmental data using interactive charts.
- Calculate AQI based on CPCB standards.
- Understand relationships between climate variables and pollution levels.

Without such a dashboard, environmental analysis becomes slow, less interactive, and difficult to interpret.

3. Proposed Solution

To overcome these challenges, the Air Quality, Pollution & Climate Analytics Dashboard provides an interactive web application developed using Streamlit.

The dashboard allows users to:

- Filter data by city and date.
- Analyze AQI trends over different time periods.
- View important environmental KPI cards.
- Explore pollutant distributions using EDA.
- Compare pollution levels across cities.
- Study the relationship between climate factors and AQI.
- Detect outliers using Box Plot analysis.
- Calculate Air Quality Index instantly using CPCB guidelines.
- Generate interactive visualizations for better understanding.

The dashboard transforms complex environmental datasets into meaningful insights through interactive analytics.

4. What is Air Quality, Pollution & Climate Analytics?

Air Quality, Pollution & Climate Analytics is a Python-based interactive dashboard designed for analyzing environmental datasets.

The application integrates Exploratory Data Analysis, statistical visualization, and CPCB AQI calculations to provide both descriptive and analytical insights.

Key Features

- Interactive Dashboard
- Dashboard Filters
- Environmental KPI Cards
- AQI Trend Analysis
- AQI Category Distribution
- Pollutant Deep Dive (EDA)
- Distribution & Density Analysis
- Box Plot Analysis
- Climate vs Pollution Analysis
- Live CPCB AQI Calculator
- Interactive Plotly Visualizations

5. Technology Stack



6. Dataset

Source: Kaggle

The dataset contains historical air quality and climate records collected from major Indian cities during 2024–2025.

The dataset includes the following attributes:

- City
- Date
- AQI
- PM2.5
- PM10
- NO₂
- SO₂
- CO
- O₃
- Temperature
- Humidity
- Wind Speed

These parameters are used for environmental analysis, pollutant profiling, and AQI visualization.

7. Dashboard Modules

The dashboard is divided into four major modules.

Climate & AQI Dashboard

Displays environmental KPI cards, AQI trends, AQI categories, and overall environmental summaries.

Pollutant Deep Dive (EDA)

Provides detailed exploratory analysis of pollutants including histograms, density plots, and statistical distributions.

Climate vs Pollution

Analyzes the relationship between weather conditions and pollution levels using scatter plots and correlation analysis.

Live CPCB AQI Calculator

Allows users to enter pollutant concentrations manually and instantly calculates the Air Quality Index according to CPCB standards with a graphical AQI gauge.

8. Dashboard Visualizations

The dashboard includes several interactive visualizations such as:

- AQI Trend Over Time
- AQI Category Distribution
- Environmental KPI Cards
- Pollutant Distribution
- Density Plot
- Box Plot Analysis
- Scatter Plot (Climate vs Pollution)
- Wind Speed vs AQI Analysis
- Live AQI Gauge
- Interactive Filters

These visualizations help users identify pollution trends, compare cities, and understand environmental conditions effectively.

9. Results

The Air Quality, Pollution & Climate Analytics Dashboard successfully demonstrates how interactive data visualization can simplify environmental analysis.

The project enables users to:

- Analyze historical AQI trends.
- Compare pollution levels across cities.
- Explore pollutant distributions.
- Detect abnormal pollution patterns.
- Study climate and pollution relationships.
- Calculate AQI instantly using CPCB standards.

The application provides a user-friendly platform for environmental monitoring and educational purposes.

10. Conclusion

Air Quality, Pollution & Climate Analytics is a practical implementation of Python-based data analytics for environmental monitoring. The dashboard combines interactive visualizations, exploratory data analysis, and CPCB AQI calculations into a single platform.

The project enhanced practical knowledge of Python, Streamlit, data visualization, exploratory data analysis, and dashboard development while demonstrating how environmental datasets can be transformed into meaningful insights for better decision-making.

12. Screenshots



